

PhyloRBT: A Phylogenetic Approach to Detect Reference Bias in

Phylogenomic Datasets

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ABSTRACT

Recent technological advancements have enabled the rapid generation of high-quality genomes across the tree of life, often resulting in multiple reference genomes for clades of phylogenetic interest. These reference genomes are often used to reconstruct phylogenetically informative loci from short-read data of newly-sequenced species. However, this approach can introduce reference bias where the reconstructed loci have erroneous similarities to those of the reference genome. Since reference bias can seriously affect downstream analyses, it is important to assess its presence in phylogenomic datasets. In this study, we propose PhyloRBT (phylogenetic reference bias test) to detect reference bias by reconstructing each locus multiple times using different reference genomes and then measuring the phylogenetic correlation between these reconstructions and the corresponding locus from the reference genomes. We applied PhyloRBT to hundreds of BUSCO loci reconstructed from short-read data of nine *Eucalyptus* species using 34 different reference genomes. Across the nine species, we found that more than a quarter of the reconstructed loci had significant evidence of reference bias. Excluding putatively biased loci from species tree inference resulted in a species tree topology that is more consistent with expectations from previous studies. In conclusion, PhyloRBT offers a

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straightforward way to detect reference bias in individual loci, and to selectively remove those biased loci from downstream analyses.

Keywords: multiple references, pairwise distance, correlation analysis, BUSCO, Eucalyptus

INTRODUCTION

Modern phylogenomic studies use hundreds or thousands of loci to infer various aspects of the evolutionary history of many taxonomic groups (Young and Gillung 2020; Lozano-Fernandez 2022). For example, He et al. (2024) used 1,764 single-copy, complete BUSCO loci (Simão et al. 2015) from 487 Basidiomycota species to infer the species tree and estimate the divergence times of the group, while Stiller et al. (2024) estimated the phylogeny of 363 Neoaves (modern birds) species using 63,430 intergenic loci. While these inferences aimed to answer different evolutionary questions using different approaches, all rely on the accuracy of individual gene sequences and gene trees being analysed. When a high-quality reference genome is not available for the species of interest, locus sequences are often reconstructed by mapping the sequencing reads of the newly-sequenced species to one or more reference genomes from closely-related taxa (e.g., Croucher et al. 2011; Bayly et al. 2013; Pease et al. 2016; Edelman et al. 2019). However, mapping can introduce *reference bias*, where the sequencing reads from the newly-sequenced species are more likely to map successfully to a reference genome when they contain variants that make them more similar to that genome, while reads with non-reference alleles might be mapped to the wrong region of the genome or not mapped at all (Degner et al. 2009; Stevenson et al. 2013; Lin et al. 2024). As a result, reconstructed sequences that are derived from the mapped reads can be erroneously similar to the reference genome. Reference bias has been shown to affect population genetics inferences (Nevado et al. 2014; Brandt et al. 2015; Gopalakrishnan et al. 2017; Shafer et al. 2017; Günther and Nettelblad

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2019), quantification of gene expression (Munger et al. 2014; Panousis et al. 2014; Zhan et al. 2021; Coombes et al. 2024), and phylogenetics inferences. For example, Valiente-Mullor et al. (2021) showed that the positions of two *Legionella pneumophila* isolates tended to cluster together with their reference genomes on the species tree, while Bertels et al. (2014) showed that using a single reference genome during mapping can result in an inaccurate tree especially when analysing highly divergent taxa. This also highlights that reference bias can be exacerbated when the reference and the newly-sequenced species are more evolutionary diverged (Gopalakrishnan et al. 2017; Bohling 2020; Huang et al. 2024). As reference bias can seriously impact downstream analyses, it is important to assess its presence in empirical datasets.

Various approaches have been developed to assess reference bias in genomic datasets. For example, AMBER measures reference bias on ancient DNA datasets based on metrics such as mapping breadth and depth (Dolenz et al. 2024). Similarly, *biastools* computes a bias score for diploid individuals based on read depth, density of the alternative alleles, and frequency of sites that are inconsistent with the diploid state (Lin et al. 2024). Another approach that has been suggested to mitigate reference bias is to use multiple reference genomes (Bertels et al. 2014; Chen et al. 2021) or pan-genome graphs (Schneeberger et al. 2009; Valenzuela et al. 2018; Kim et al. 2019; Sirén et al. 2021) from closely-related species during mapping. This is becoming increasingly common in phylogenomic studies because long read sequencing has enabled *de novo* assembly of high-quality reference genomes for many groups, including non-model organisms (Ekblom and Wolf 2014; Sedlazeck et al. 2018; Rhie et al. 2021). In principle, the availability of multiple reference genomes should minimise the effects of reference bias, because they should collectively better represent the allelic variation in the newly-sequenced species. It also presents an opportunity to measure reference bias in new ways.

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77 In this study, we propose PhyloRBT (phylogenetic reference bias test), a new approach that
78 leverages the availability of multiple reference genomes to measure reference bias in
79 individually reconstructed loci from short-read data. PhyloRBT works by first mapping the
80 short reads from a newly-sequenced species to multiple reference genomes. Then, we
81 reconstruct each locus of interest once for each reference genome. In an ideal world (i.e., when
82 there is no reference bias or other errors in the reconstruction), the sequence of each individual
83 locus should be identical no matter which reference genome was used during its reconstruction.
84 However, in the presence of reference bias, the sequence of an individual locus will tend to be
85 more similar to the same locus in the reference genome used to reconstruct it than would be
86 expected by chance. We leverage this difference using statistical tests to detect reference bias.
87 In this study, we focus on BUSCO (Benchmarking Universal Single Copy Orthologs; Simão
88 et al. 2015; Manni et al. 2021) as our target loci because they are widely-used for phylogenomic
89 inference (e.g., Kanzi et al. 2020; Li et al. 2021; Ferguson et al. 2024a) and expected to be
90 present in single copies in most genomes, which minimalises the effects of paralogs in our
91 analyses.

92

93 The BUSCO pipeline works by first extracting the list of BUSCO from the taxonomic group
94 of interest from OrthoDB (Zdobnov et al. 2021). Then, it uses tBLASTn (Camacho et al. 2009)
95 to identify and retrieve the genomic coordinates of the BUSCO loci on the input genome
96 assembly, and uses a gene predictor tool such as AUGUSTUS (Stanke et al. 2004) or MetaEuk
97 (Levy Karin et al. 2020) to predict the gene structure (e.g., exon and intron boundaries) for
98 each extracted BUSCO. Lastly, the pipeline uses HMMER 3 (Eddy 2011) to evaluate the
99 sequences of the extracted BUSCO against the HMM profiles from the database and classify
100 those BUSCO as either complete (single-copy or duplicated), fragmented, or missing from the

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input assembly. When researchers have the short-read data from a newly-sequenced species, they need to first assemble the reads before reconstructing the BUSCO loci, which can be done in two ways. One option is to do *de novo* assembly of the short-read data. This avoids reference bias altogether because no reference genomes are used during the assembly. However, *de novo* assembly from exclusively short reads can be challenging, as these assemblies tend to be more fragmented and contain a higher proportion of gaps compared to those built from long sequencing reads (Paszkievicz and Studholme 2010; Liao et al. 2019). An alternative and commonly-used approach involves mapping the short reads to a closely-related reference genome, followed by variant calling to generate a consensus sequence. This method is susceptible to reference bias as the reconstructed BUSCO sequences may be affected by how the short sequencing reads align to the reference genome. Here, we focus on the second approach (i.e., mapping) to infer BUSCO loci from short-read datasets and assess how reference bias may affect their sequence reconstruction.

In this study, we first describe PhyloRBT to detect reference bias in phylogenomic datasets. We then apply PhyloRBT to assess the extent of reference bias in an empirical dataset from Eucalypts (*Eucalyptus*, *Corymbia*, and *Angophora*), comprising recently-assembled reference genomes and the associated short sequencing reads from 35 species (Ferguson et al. 2024a). To test our method, we selected the short-read data from nine *Eucalyptus* species that are spread across the species tree of the group (Ferguson et al. 2024a) and individually mapped each to all 34 taxa in the dataset that are not from the same species as the short-read data. We show that statistically significant evidence of reference bias is found in a large proportion of the reconstructed BUSCO loci across species, and describe how this can affect the downstream phylogenomic inferences.

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MATERIALS AND METHODS

Overview of PhyloRBT

In order to measure the extent of reference bias in phylogenomic datasets, we designed PhyloRBT that leverages the availability of multiple reference genomes and focuses on one locus at a time. All the code necessary to reproduce the methods in this paper is available at <https://github.com/jeremiasivan/PhyloRBT>.

PhyloRBT works by using each reference genome in turn to reconstruct each locus for each set of sequencing reads. For example, if there are three reference genomes available for the clade of interest, we would reconstruct each locus three times (once for each reference genome) for each newly-sequenced species in our dataset (where each new species is represented by a single set of short sequencing reads). If there is no reference bias and no error in the reconstructions, the three reconstructed loci from the same set of sequencing reads would be identical to each other. In contrast, reference bias would make the three reconstructed sequences more similar to the reference genomes than we would expect by chance. This pattern is different from the stochastic error that may occur during the mapping and variant calling processes, which would make the three reconstructed locus sequences different from one another independently of the reference genome used to reconstruct them. PhyloRBT distinguishes reference bias from stochastic error by building a phylogenetic tree of all reconstructed versions of the locus of interest along with the sequences of the same locus extracted from the reference genomes. We then ask if the *reconstructed* loci are more similar to their corresponding *reference* loci than would be expected by chance, indicating evidence of reference bias.

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Figure 1 gives an example where we are interested in reconstructing a single locus from a single set of short reads (*SI*), and we have three reference genomes available from the clade of interest (*RefA*, *RefB*, *RefC*). First, we reconstruct three versions of the respective locus for our target species (*SI*) by mapping the short reads to each of the three reference genomes. We then build a locus tree from the three versions of the reconstructed locus, as well as the locus sequences from the three reference genomes, resulting in a tree with 6 tips. In a situation without reference bias, the three reconstructed loci from *SI* would be identical to one another, and the locus tree would group the three reconstructed loci together (i.e., *SI--RefA*, *SI--RefB*, and *SI--RefC* are grouped together on Fig. 1a). In the presence of reference bias, the reconstructed locus from *SI* would cluster with the locus sequence from the reference genome used to reconstruct it (i.e., *SI--RefA* groups with *RefA*, *SI--RefB* groups with *RefB*, and *SI--RefC* groups with *RefC* on Fig. 1b). The stronger the effect of reference bias, the stronger the clustering of the reconstructed loci with their respective reference loci.

PhyloRBT tests for the presence of reference bias on a given locus by comparing the phylogenetic distances of the reconstructed loci to their respective reference loci. In order to do this, we start by randomly choosing a single sequence from the reference genomes as the focal sequence to avoid pseudoreplication (e.g., *RefA* on Fig. 1). Then, we measure the pairwise phylogenetic distance of each reference locus to the focal tip (i.e., reference phylogenetic distance, x-axis on Fig. 1), as well as the pairwise phylogenetic distance of each reconstructed locus to the focal tip (i.e., reads phylogenetic distance, y-axis on Fig. 1). We use a linear model and a Spearman's rank-order correlation to ask if the two measures of pairwise phylogenetic distances are more correlated than would be expected by chance (Schober et al. 2018). Significant results ($p < 0.05$) from either or both tests would suggest the presence of reference

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bias for a single locus reconstructed from a single set of short reads that are mapped to multiple references.

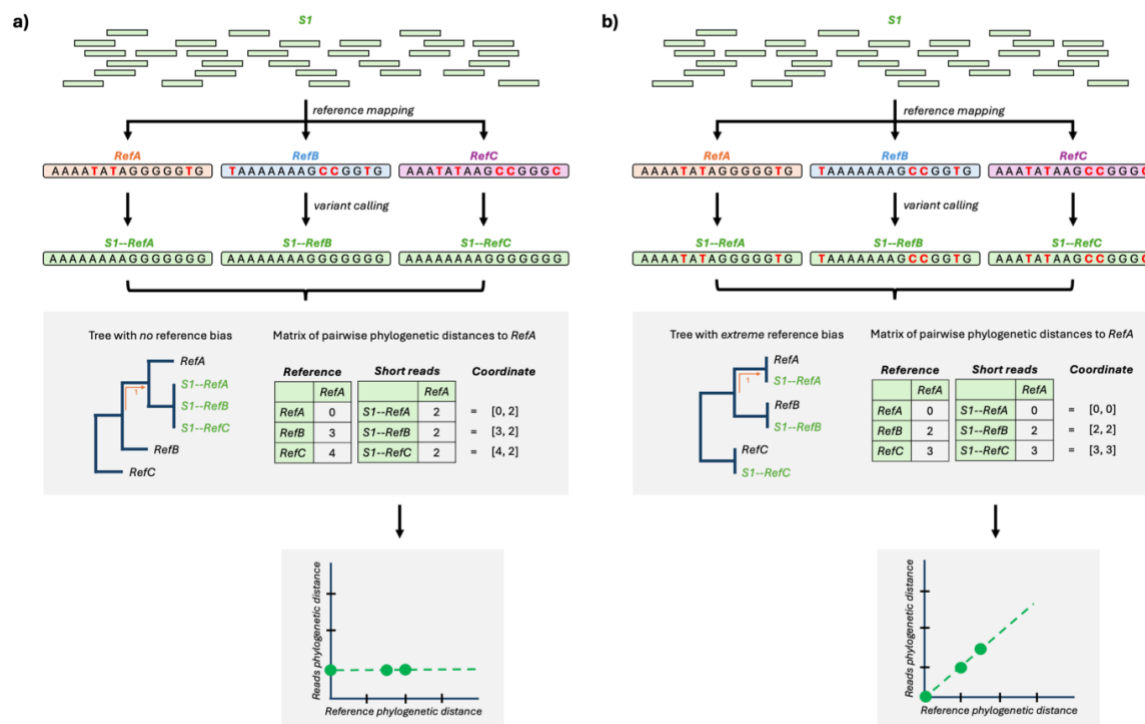


Figure 1. Overview of PhyloRBT when there is (a) no reference bias, and (b) total reference bias. Pairwise phylogenetic distance refers to the pairwise distance in branch length unit. *SI* denotes a single set of short reads from a previously-unsequenced species, while the *SI--RefSX* notation refers to the short reads from *SI* that are mapped to the reference genome from species *X*. Colouring is based on the species.

Application to an Empirical Dataset

Overview

We applied PhyloRBT to an empirical dataset of 35 *Eucalypts* (comprising *Eucalyptus*, *Corymbia*, and *Angophora*), where each species has its own well-polished *de novo* reference genome and corresponding set of short reads (Ferguson et al. 2024a). We used short-read data from nine representative *Eucalyptus* species. Then, we ran the BUSCO pipeline to extract

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BUSCO loci from each set of short reads, using each of the 35 reference loci in turn. This results in 35 reconstructions of each BUSCO locus for each of the 9 short-read species in our dataset. We then used PhyloRBT to test for reference bias in each BUSCO locus from each species, and assessed whether removing BUSCO loci with evidence of reference bias affected the species tree inference of the group.

Datasets

Our dataset comprises 35 taxa from Eucalypts: 32 *Eucalyptus*, 2 *Corymbia* (*C. maculata*, *C. calophylla*), and one *Angophora* (*A. floribunda*) (Ferguson et al. 2024a). We selected nine representative *Eucalyptus* taxa to test our method – this should be sufficient to effectively test our new approach and evaluate the scale of reference bias in a dataset, but small enough to ensure computational efficiency. The nine representative taxa represent three groups of three closely-related species that are spread across the species tree from Ferguson et al. (2024a): *E. curtisii*, *E. erythrocorys*, and *E. tenuipes* in group one; *E. grandis*, *E. globulus*, and *E. viminalis* in group two, and *E. albens*, *E. melliodora*, and *E. sideroxylon* in group three (Fig. S1). This configuration allowed us to test the presence of reference bias in BUSCO inference when a set of short reads was mapped to a reference coming from closely-related taxa (e.g., *E. curtisii* short reads mapped to *E. tenuipes* reference genome), distantly-related taxa (e.g., *E. curtisii* short reads mapped to *E. albens* reference genome), and different genera (e.g., *E. curtisii* short reads mapped to *A. floribunda* reference genome). To avoid circularity, short reads were never mapped to the genome from the same species from which they were derived, see below.

Reference Genomes: Downloading Data and Extracting Reference BUSCO

We downloaded the reference genomes of the 35 Eucalypts species from NCBI (BioProject PRJNA509734) using NCBI Datasets (<https://www.ncbi.nlm.nih.gov/datasets/>) based on their

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accession numbers. In order to extract BUSCO from the reference genomes, we ran BUSCO pipeline v5.8.0 (Manni et al. 2021) on each reference genome using *eudicots_odb10* as lineage and MetaEuk (Levy Karin et al. 2020) as the gene predictor tool. We chose *eudicots* because it is the most specific taxonomic group that encompasses *Eucalypts*, while MetaEuk provides a balance between speed (i.e., it is faster than Augustus (Stanke et al. 2004)) and accuracy (i.e., it is slower than miniprot (Li 2023) but less sensitive when the assemblies are highly-divergent from the database). We then retrieved all BUSCO that were identified as single-copy and complete for each of the 35 reference genomes, excluding duplicated and fragmented BUSCO to ensure that the subsequent phylogenetic inferences were not affected by paralogs. The code to run this analysis are available at https://github.com/jeremiasivan/PhyloRBT/data_preparation/2_extract_busco/2_refseq.Rmd.

Short Reads

Downloading raw data, quality control, and mapping

We downloaded the raw short sequencing reads of the nine representative *Eucalyptus* taxa mentioned above from NCBI (BioProject PRJNA509734) using SRA Toolkit (<https://github.com/ncbi/sra-tools>) based on their accession numbers, consisting of two files per taxon: forward and reverse reads. Then, we performed quality control on each set of short reads using AdapterRemoval v2 (Schubert et al. 2016) by trimming the adapters and filtering out low-quality reads ($Q < 25$) as well as leading and trailing ‘N’ using the command: `AdapterRemoval --trimqualities ---trimns --adapter-list file_adapters --minquality 25`.

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In order to mimic an empirical scenario in which a reference genome from the same species as the newly-sequenced species is unavailable, we mapped the short-read data of each of the nine representative *Eucalyptus* species to all reference genomes except the one from the same species. To do this, we used BWA-MEM2 (Li and Durbin 2009; Vasimuddin et al. 2019) with default settings, resulting in 34 SAM files per representative species. We compressed the SAM files to BAM format and sorted the mapped reads based on their genomic coordinates using SAMtools v1.18 (Li et al. 2009). Then, we removed putative PCR duplicates using SAMtools `rmdup` (Li et al. 2009) and performed variant calling using `mpileup` command from BCFtools v1.19 (Li 2011). Lastly, we filtered out low-quality variants ($QUAL \leq 15$ & $MAPQ \leq 30$) (Ferguson et al. 2024b), did variant normalisation, and generated the consensus sequence for each BAM file using BCFtools `view`, `norm`, and `consensus`, respectively, totaling 306 consensus sequences. The code to run this analysis are available at https://github.com/jeremiasivan/PhyloRBT/data_preparation/1_readmap/.

Extracting

BUSCO

In order to extract BUSCO from the consensus sequences, we ran the BUSCO pipeline on each consensus sequence with the same parameters as the ones used for the reference genomes. Then, for each of the nine representative species, we retrieved all BUSCO that were found as single-copy and complete in *all* versions of the consensus sequences – representing different reference genomes used during mapping – and *all* 34 reference genomes being used. Moreover, in order to ensure that each reconstructed BUSCO from the consensus sequence is well-supported by the short reads (i.e., if the BUSCO has zero coverage on the consensus sequence, then the reconstructed BUSCO would only reflect the sequence of the reference genome), we measured the coverage of each BUSCO on each consensus sequence by first converting the BUSCO GFF files outputted by the BUSCO pipeline to a BED format using BEDOPS v2.4.41

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(Neph et al. 2012). Then, we calculated the coverage of each BUSCO on each consensus sequence using Samtools `depth` (Li et al. 2009). We set the minimum coverage for each BUSCO to be 10x for *all* consensus sequences (because the mean sequencing depth of the short reads is ~20x), while the maximum is two times the mean coverage of the mapped reads (Table S1). These are strict filters that would not be necessary for a typical phylogenomic analysis. However, they are helpful to accurately assess the scale of reference bias in the dataset because every reconstructed locus is present and well-supported for every species, ensuring that missing data is not affecting our power to detect reference bias in this study. The code to run this analysis are available at https://github.com/jeremiasivan/PhyloRBT/data_preparation/2_extract_busco/3_shortreads.Rmd.

Generating BUSCO Alignments and Trees

For each BUSCO from each representative species, we combined the amino acid sequences from all reference genomes and consensus sequences into one FASTA file (i.e., each FASTA file contained 34 reconstructed versions of the locus from the representative species, as well as the 34 versions of the same locus that were extracted directly from the reference genomes). We then ran MAFFT v7.520 (Katoh et al. 2002) with default settings (FFT-NS-2) to align these 68 sequences. In order to assess the effect of filtering out poorly-aligned regions from the alignments on reference bias, we generated another sets of BUSCO alignments where each alignment was filtered using the `-automated1` flag from TrimAl v1.2rev59 (Capella-Gutiérrez et al. 2009). We then built phylogenies for the trimmed and untrimmed alignments using IQ-TREE2 (Minh et al. 2020b) with 1,000 UFBoot replicates (Hoang et al. 2018) and the best model from ModelFinder (Kalyaanamoorthy et al. 2017), resulting in two gene trees (based on the filtered and unfiltered alignments) for each BUSCO locus. The code to run this

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analysis are available at https://github.com/jeremiasivan/PhyloRBT/data_preparation/2_extract_busco/4_gene_trees.Rmd.

Measuring Reference Bias

We ran PhyloRBT on individual BUSCO trees to detect reference bias on each tree (from both the trimmed and untrimmed alignments). PhyloRBT first calculates the pairwise phylogenetic distances between all the reference genomes and consensus sequences using `cophenetic.phylo` function in *ape* (Paradis et al. 2004). In order to measure reference bias for each set of short reads in each tree, we arbitrarily selected the locus from the reference genome of *E. cloeziana* as our focal sequence (note that the selection of focal species should not affect the results; Fig. 1), then extracted the pairwise distances from the focal sequence to each reference genome and to each reconstructed sequence of the representative species (see Fig. 1 for reference). Finally, we ran a linear model and a Spearman's rank order correlation to test for significant associations between the two sets of distances using R (R Core Team 2023), and counted the proportion of BUSCO loci with statistically significant results ($p < 0.05$) for each test independently, and the proportion when both tests were significant. The code to run this analysis are available at https://github.com/jeremiasivan/PhyloRBT/codes/2_run.Rmd.

Assessing the Effect of Reference Bias using Concordance Factors

In the absence of reference bias, *all* reconstructions of the same sample (or species) should cluster together on the tree (Fig. 1). In order to assess the utility of PhyloRBT to detect putatively biased loci and improve the clustering of the species reconstructions by excluding such loci, we inferred ASTRAL trees (Zhang et al. 2018) for each of the nine representative species (with the 34 reference genomes) based on three sets of BUSCO gene trees: (i) all BUSCO loci, (ii) BUSCO with significant evidence of reference bias from *either* or both

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statistical tests, and (iii) BUSCO with *no* evidence of reference bias based on the statistical tests. We hypothesise that the clustering of the newly-sequenced species would have higher concordance factors (CFs) on ASTRAL tree that was constructed only from putatively unbiased loci compared to when using all loci.

For each of the 27 ASTRAL trees (i.e., 9 representative species * 3 sets of BUSCO trees), we retrieved the ASTRAL posterior probability of the branch that unites *all* 34 reconstructed loci from the newly-sequenced species. We also retrieved the ASTRAL quartet score (qCF) and computed the gCF (gene concordance factor) and sCF (site concordance factor) of the same branch as these metrics are designed to measure the variation of tree topologies within the data (i.e., how many quartets, genes, and individual sites support that branch) (Minh et al. 2020a; Mo et al. 2023; Lanfear and Hahn 2024). We computed the gCF and sCF of each ASTRAL tree using IQ-TREE2 (Minh et al. 2020b) with the following commands:

```
iqtree2 -t astral.tre --gcf busco_trees.tre
iqtree2 -te astral.tre -p dir_busco_alignments --scf 100
```

where `astral.tre` reflects the ASTRAL tree, `busco_trees.tre` refers to a file containing the list of BUSCO trees in Newick format with one tree per line, and `dir_busco_alignments` refers to the directory where individual BUSCO alignments are stored. The code to run this analysis are available at https://github.com/jeremiasivan/PhyloRBT/codes/3_topology_test.Rmd.

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Assessing the Effect of Reference Bias on Species Tree Inference

While including all reconstructions of the same species on the phylogeny is informative to measure the support of the clustering, most phylogenomic studies would only use one reconstruction of the newly-sequenced species on the species tree. In order to assess the effect of reference bias on individual reconstructions, we first selected one species from each group (*E. albens*, *E. erythrocorysi*, and *E. grandis*; Fig. S1). For each representative species, we then built ASTRAL trees that consists of one reconstruction of the species and the 34 reference genomes using two sets of BUSCO loci: *all* BUSCO and only putatively unbiased BUSCO, totaling in 204 ASTRAL trees (3 species * 34 reconstructions * 2 sets of BUSCO trees). However, we constrained the tree topology of the 34 reference genomes using the published tree (Ferguson et al. 2024a) in ASTRAL v5.6.9 (Rabiee and Mirarab 2020), which allows us to compare the placement of the newly-sequenced species on the tree when mapped to different reference genomes.

For each reconstruction, we then assessed if excluding the putatively biased loci from the ASTRAL tree inference improved the consistency of the tree compared to the published tree (Ferguson et al. 2024a). We measured this by calculating the phylogenetic distance (in coalescent unit) of the species placement between the two trees – namely how many coalescent units does the species need to traverse moving from its placement on the ASTRAL tree to reach its placement (i.e., branch) on the published tree. We hypothesise that ASTRAL tree inferred from only putatively unbiased BUSCO would have lower phylogenetic distance (to the published tree) than the one inferred from all BUSCO. The code to run this analysis are available at https://github.com/jeremiasivan/PhyloRBT/codes/3_topology_test.Rmd.

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RESULTS

Reference bias is pervasive in phylogenomic datasets

Across the nine representative species, the average coverage of mapped short reads ranged from 8.214x (for *E. tenuipes* mapped to *C. maculata*) to 37.564x (for *E. sideroxylon* mapped to *E. tenuipes*), with generally lower coverage and percentage of mapped reads when using *Corymbia* and *Angophora* reference genomes (Table S1-S2). For each of the nine short-read datasets, our analyses resulted in the reconstruction of between 192 and 466 BUSCO loci (Table S3). Based on *both* the linear model and Spearman's rank order correlation test, approximately 42-49% (average 45.53%) of these loci show significant evidence of reference bias across the representative species (Fig. 2 (left) and Table S3). This proportion is even higher when considering individual tests: 66-75% (average 70.11%) based on the linear model only, and 48-57% (average 51.68%) based on the Spearman's rank order correlation only (Fig. 2 (left), Table S3). When excluding the distantly-related *Angophora* and *Corymbia* reference genomes from the analyses, the proportion of BUSCO loci with significant results for both statistical tests decreases to 25-31% (average 28.23%) for both tests combined; 46-54% (average 50.66%) for only linear model; and 32-37% (average 34.58%) for only Spearman's rank correlation (Fig. 2 (right), Table S3). Filtering out poorly-aligned regions made negligible and inconsistent changes to the proportion of BUSCO loci with significant evidence of reference bias across representative species (Table S4), so for brevity we do not discuss it further here.

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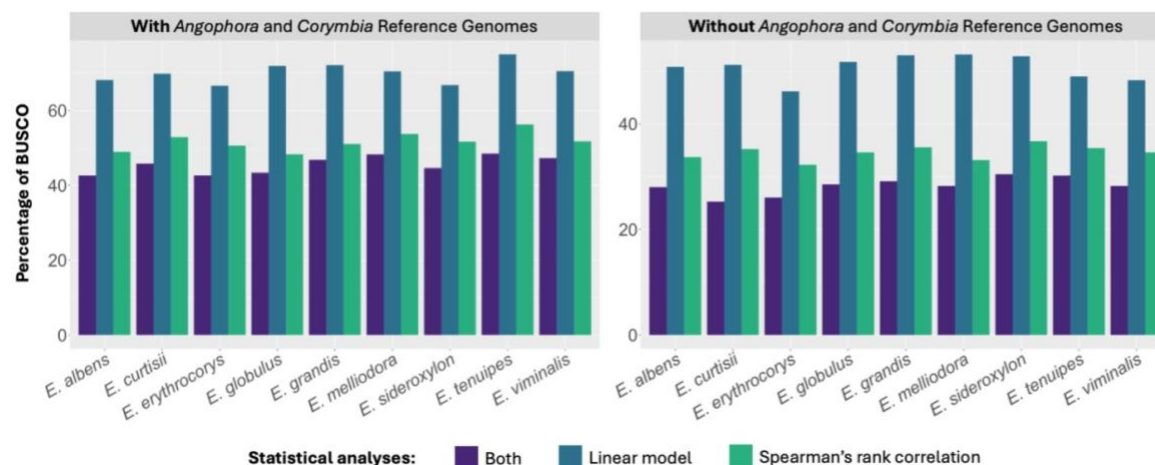
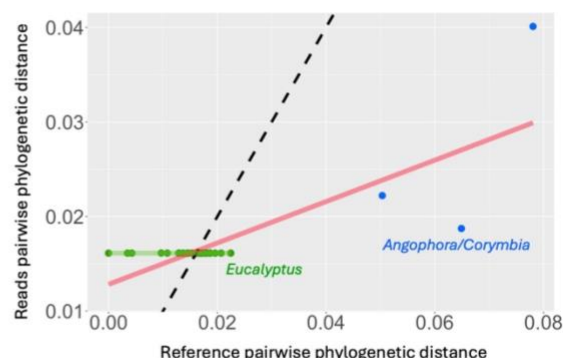
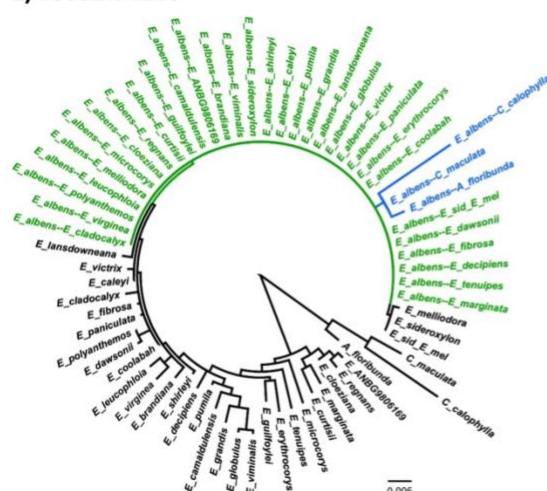


Figure 2. Percentage of BUSCO with significant results from linear model and/or Spearman's rank order correlation test ($p<0.05$) across nine representative *Eucalyptus* species. Colouring denotes the statistical tests being used.

Figure 3 shows an example of two BUSCO loci that were reconstructed from *E. albens* short-read data. For BUSCO 4838at71240, all reconstructed BUSCO from *E. albens* (except three that were mapped to *Angophora* and *Corymbia* references) are clustered together with zero phylogenetic distance (Fig. 3a), reflecting no reference bias. This is reflected by the results from the linear model; when we excluded the reconstructed BUSCO that were mapped to *Angophora* and *Corymbia* reference genomes, the correlation between the reads and the reference pairwise distances is not statistically significant ($R^2=0$, $p=0.182$). On the other hand, BUSCO 31772at71240 shows evidence of reference bias where some reconstructed loci of *E. albens* appear in the tree very close to the reference genomes that were used to reconstruct them (Fig. 3b). This observation is supported by the linear model, which shows significant correlations between the two distances both when the reconstructed BUSCO that were mapped to *Angophora* and *Corymbia* reference genomes are included ($R^2=0.857$, $p<0.001$), and when the *Angophora* and *Corymbia* reference genomes are excluded ($R^2=0.649$, $p<0.001$).

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a) 4838at71240



b) 31772at71240

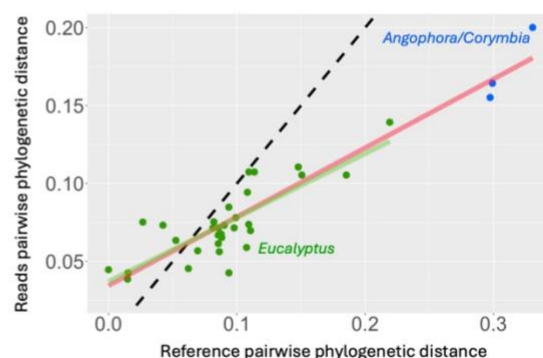
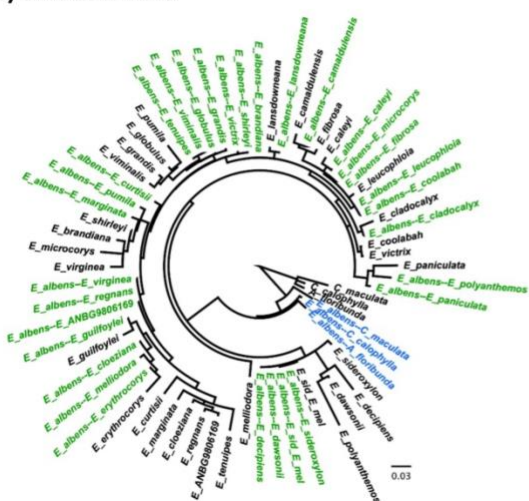


Figure 3. BUSCO (a) 4838at71240 and (b) 31772at71240 for *E. albens* consensus sequences. Left: BUSCO trees; right: scatter plots of the BUSCO pairwise distances to the focal species (i.e., *E. cloeziana*). Dashed line shows $x = y$ line; red solid line shows the fitted line based on linear model with *Angophora* and *Corymbia* reference genomes; green solid line shows the fitted line based on linear model without *Angophora* and *Corymbia* reference genomes. Green-coloured tips denote *E. albens* consensus sequences with *Eucalyptus* reference genomes, while the blue-coloured tips denote *E. albens* consensus sequences with either *Corymbia* or *Angophora* reference genomes.

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Excluding putatively biased loci increases the concordance factors of the species clustering

Excluding the putatively biased loci (i.e., BUSCO loci with a significant result from either or both statistical tests) generally increases the concordance factors of the reconstructions clustering on the ASTRAL tree compared to when using all BUSCO loci. Specifically, using only the putatively unbiased loci increases the gCF and qCF of the relevant branch by on average 6.11% and 6.08%, respectively, except for *E. globulus* (with 1.04% gCF and 14.25% qCF decrease) and *E. albens* (with 0.57% qCF decrease) and *E. albens* (with 0.57% qCF decrease) (Fig. 4, S2-S10). However, the changes in sCF are more inconsistent, where it increases by 58.05% for *E. tenuipes* when using only putatively unbiased loci, but decreases by 54.15% for *E. curtisii* (Fig. 4, S2-S10).

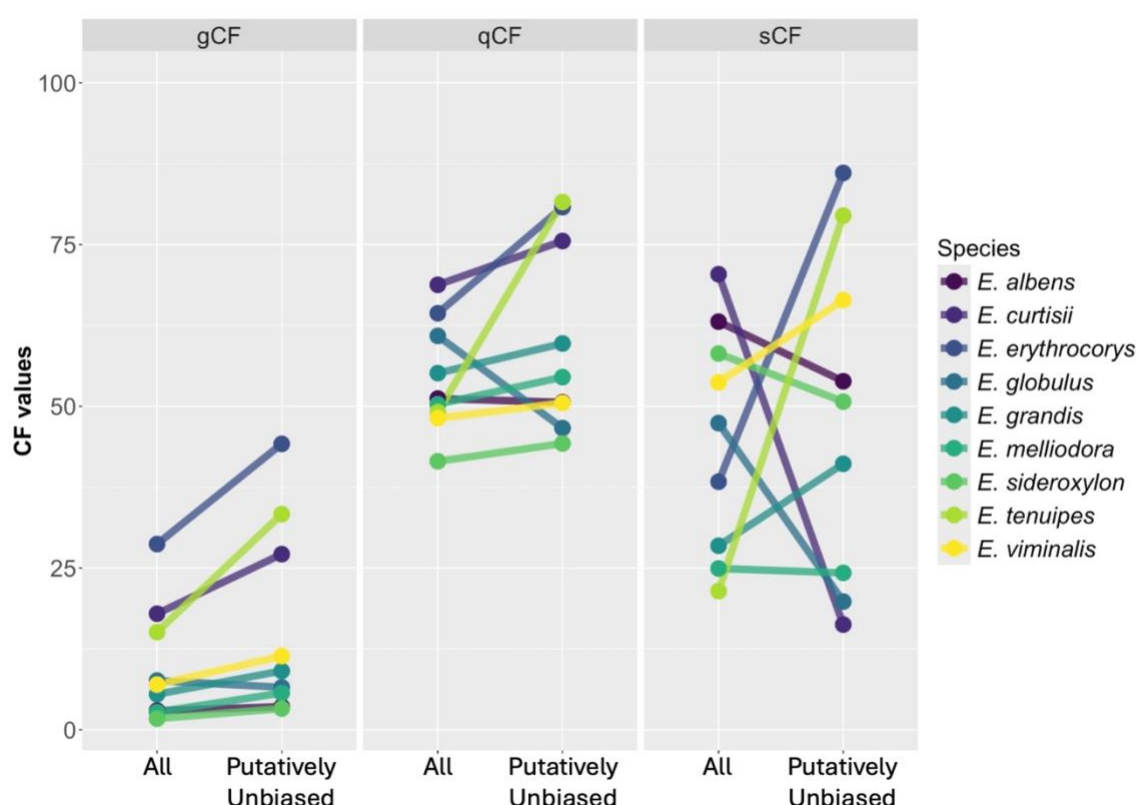


Figure 4. Concordance factors of the relevant branch on the ASTRAL trees inferred using *all* BUSCO loci and only putatively unbiased BUSCO loci. Relevant branch refers to the branch that leads to all reconstructed versions of the species. gCF: gene concordance factors; qCF: quartet concordance factor; sCF: site concordance factor. Colouring denotes different species.

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Reference bias affects the species placement on the species tree

When only one reconstruction was included in the species tree inference, the placement of the newly-sequenced species varies depending on the reference genome being used. For example, Figure S11-S14 show that the position of *E. albens* reconstruction shifts within the *Adnataria* clade, often with >0.95 posterior probability support even when the placement is inconsistent with the published tree (Ferguson et al. 2024a). This pattern is further supported by calculating the phylogenetic distance between the ASTRAL tree and the published tree. Across the three representative species, Figure 5 (*All*) shows that the newly-sequenced species is placed much deeper in the tree when using reference genomes from *Corymbia* or *Angophora* (except for the short reads of *E. erythrocorys* that is placed at the crown of *Eucalyptus* in the published tree). Conversely, using reference genomes from closely-related taxa pulls the reconstruction closer to the reference genome itself (except for the short reads of *E. albens*; light green-coloured dots in Fig. 5 (*All*)). Excluding the putatively-biased BUSCO loci does not consistently reduce the phylogenetic distance to the published tree (Fig. 5 (*Putatively Unbiased*)). However, when we used *both* reference genome from closely-related taxa during mapping *and* only putatively-unbiased BUSCO during the species tree inference, the ASTRAL tree topology is always consistent with the published tree across the three representative species (light green-coloured dots in Fig. 5 (*Putatively Unbiased*)). These results further support the use of closely-related reference genome during mapping, and highlight the importance of addressing reference bias in phylogenomic analyses.

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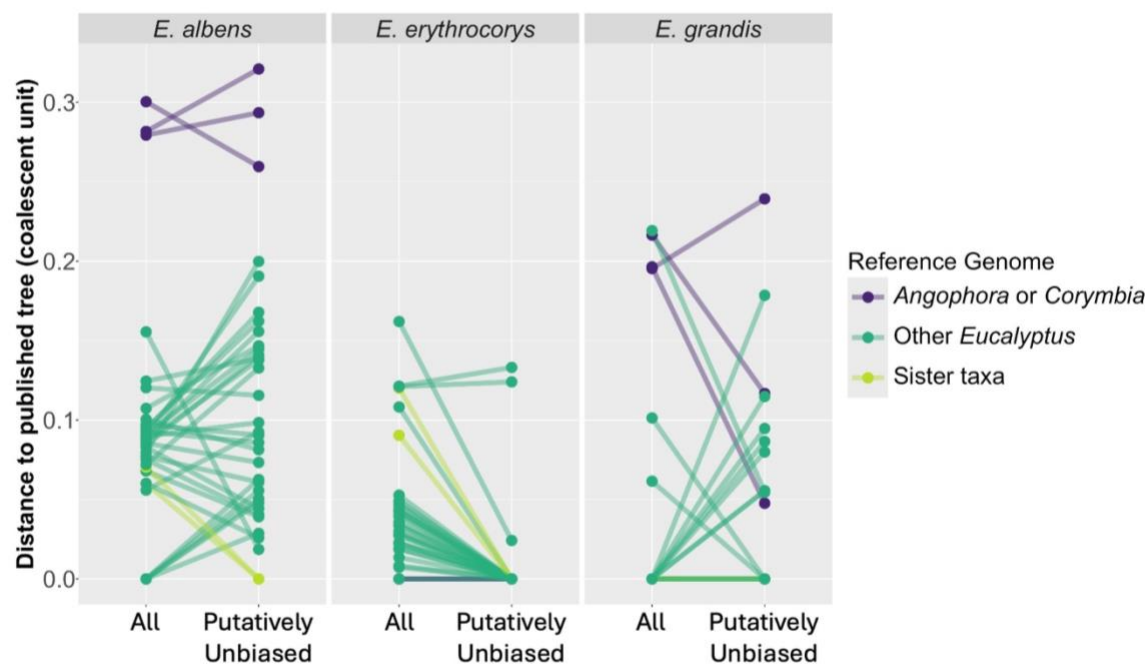


Figure 5. Phylogenetic distance between the position of three *Eucalyptus* species on the ASTRAL trees (inferred using *all* BUSCO loci and only putatively unbiased BUSCO loci) and the published tree (Ferguson et al. 2024a). The y-axis reflects the total branch lengths (in coalescent unit) needed for the reconstruction to reach the ‘true’ branch on the published tree (see red-coloured edges on Fig. S11-S14 for reference). Colouring denotes different reference genomes used for mapping (Fig. S1). Lines connect reconstruction with the same reference genome.

DISCUSSION

Many phylogenomic studies start by mapping the short-read data of newly-sequenced species to one or more high-quality reference genomes, in order to reconstruct loci for phylogenetic inference. However, reference genome selection has been found to introduce bias that can make the reconstructed locus erroneously similar to reference genome (Degner et al. 2009; Stevenson et al. 2013; Lin et al. 2024). Using the dataset of Ferguson et al. (2024a), we found that across hundreds of BUSCO loci, roughly half showed significant evidence of reference bias across nine *Eucalyptus* species based on PhyloRBT, which was exacerbated when the reference genome became more distantly-related with the newly-sequenced species (Fig. 2; Table S3).

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This highlights the risk of introducing reference bias in phylogenomic datasets generated using reference mapping, even with closely-related species.

In order to address the bias, the simplest way is to avoid it altogether by assembling the genome *de novo*, which is becoming increasingly affordable with the advancement of long-read sequencing technologies (Ekblom and Wolf 2014; Sedlazeck et al. 2018; Rhie et al. 2021). However, in cases where short sequencing reads are the only data available, *de novo* assembly is particularly challenging (Paszkiiewicz and Studholme 2010; Liao et al. 2019), even though previous studies had found success in generating draft genome assemblies from DNA extracted from fresh tissue and museum specimens, as well as ancient DNA (e.g., Li et al. 2010; Feigin et al. 2018; Varga et al. 2025). When *de novo* assembly is not feasible and mapping to a reference genome is a necessary part of a phylogenomic study, careful consideration needs to be exercised when performing data analyses in the presence of reference bias.

When we used all reconstructions of each of the nine representative *Eucalyptus* species, we found that removing the putatively biased loci from the ASTRAL tree inference improved the gCF and qCF (but not sCF) of the reconstruction clustering compared to when using *all* BUSCO loci (Fig. 4). Different to posterior probability that measures sampling variance – which tends to be small in phylogenomic datasets due to the large number of loci being used (Gadagkar et al. 2005) – concordance factors (CFs) measure the predictive ability of the species tree for any given genes (gCF and qCF) or sites (sCF) (Lanfear and Hahn 2024). As PhyloRBT detects reference bias in individual loci, excluding the putatively-biased loci reduced the degree of reference bias in the datasets and improved the reconstruction clustering, which was reflected by increased proportion of loci (gCF) and quartets (qCF) supporting the clustering. However, a single locus may contain multiple evolutionary histories due to recombination,

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such that some sites support the reconstruction clustering while others do not (Degnan and Rosenberg 2009; Mallet et al. 2016; Bryant and Hahn 2020). PhyloRBT does not account for these intra-locus recombination events because we only inferred one history (i.e., tree) per locus. Consequently, some sites within the putatively-biased loci may in fact be unbiased but were nonetheless excluded from the analysis, and vice versa. As the proportion of sites (sCF) that support the clustering did not always increase after excluding the putatively-biased BUSCO loci from the ASTRAL tree inference (Fig. 4), it suggests that intra-locus recombination is prevalent in our dataset.

Similar to previous findings (Bertels et al. 2014; Valiente-Mullor et al. 2021), we also found that reference bias affected the species tree inference of three representative species (*E. albens*, *E. erythrocorys*, and *E. grandis*), where the placement of the species on the ASTRAL tree was highly dependent on the reference genome being used during mapping (Fig. 5). For example, using the reference genomes from *Corymbia* or *Angophora* placed the *E. albens* reconstruction at the crown of *Adnataria* clade with high posterior probability support (Fig. S11). Conversely, when the reference genomes came from closely-related taxa (e.g., *E. melliodora* and *E. sideroxylon*), the *E. albens* reconstruction clustered together with the reference genome itself (Fig. S12). This highlights that reference bias can significantly affect species tree inference, often with high support value, even when the reference genome being used comes from closely-related taxa.

However, we also showed that when we used the reference genome from closely-related taxa and excluded putatively-biased BUSCO from the species tree inference, the ASTRAL tree topology became more consistent with the published tree (Ferguson et al. 2024a). Thus, we have two recommendations to address reference bias in phylogenomic datasets when multiple

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reference genomes are available. First, we should run PhyloRBT to detect loci with strong evidence of reference bias and exclude them from the species tree inference. Second, for each of the remaining loci, we should select the reconstruction that was mapped to the closest reference genome (which can be estimated directly from the short sequencing reads using tools such as Mash (Ondov et al. 2016) in the absence of a published species tree). Alternatively, we could reconstruct the remaining loci directly from the short reads of the newly-sequenced species using HybPiper (Johnson et al. 2016) or other similar tools that allow for iterative mapping during locus reconstructions (Hahn et al. 2013). In brief, iterative mapping involves: (i) mapping short reads to the reference locus, (ii) constructing a new consensus sequence from the mapped reads, and (iii) using this consensus sequence as the updated target locus. By repeating these steps until convergence (i.e., until no new reads are mapped), the reconstructed locus should be less affected by reference bias and more accurately reflect the sequence of the newly-sequenced species.

Overall, our study shows one example of the prevalence of reference bias in phylogenomic datasets and thus proposes PhyloRBT to detect the bias on individual locus using multiple reference genomes. We achieved this by individually mapping each set of short reads to multiple reference genomes, reconstructing the locus of interest from the consensus sequences, and then running correlation analyses between the reads pairwise distance (i.e., the pairwise phylogenetic distance between the consensus sequence with the focal species) and the reference pairwise distance (i.e., the pairwise phylogenetic distance between the reference genome and the focal species). While we only presented one use case of our method – namely, to infer species tree using BUSCO loci – it can also be used for other phylogenomic analyses (e.g., inference of gene trees), or even extended to evaluate different approaches that aim to mitigate reference bias during upstream analyses (e.g., selection of variant calling parameters (Shafer

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et al. 2017; Barbitoff et al. 2022; Rick et al. 2024), realignment of reads (Homer and Nelson 2010; Pirooznia et al. 2014; Mose et al. 2019), iterative mapping (Dutilh et al. 2009; Tsai et al. 2010; Hahn et al. 2013), and the use of pangenomes (The Computational Pan-Genomics Consortium 2018; Valenzuela et al. 2018; Chen et al. 2021)). In conclusion, we believe that PhyloRBT provides a valuable resource to detect reference bias in empirical datasets using multiple reference genomes, which can help to reduce its impact on downstream analyses, particularly phylogenetic inference.

SUPPLEMENTARY MATERIALS

Supplementary material will be available online upon the acceptance of the paper.

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REFERENCES

Barbitoff Y.A., Abasov R., Tvorogova V.E., Glotov A.S., Predeus A.V. 2022. Systematic benchmark of state-of-the-art variant calling pipelines identifies major factors affecting accuracy of coding sequence variant discovery. BMC Genomics. 23:155.

Ivan et al.

567 Bayly M.J., Rigault P., Spokevicius A., Ladiges P.Y., Ades P.K., Anderson C., Bossinger G.,
568 Merchant A., Udovicic F., Woodrow I.E., Tibbits J. 2013. Chloroplast genome analysis
569 of Australian eucalypts – *Eucalyptus*, *Corymbia*, *Angophora*, *Allosyncarpia* and
570 *Stockwellia* (Myrtaceae). *Molecular Phylogenetics and Evolution*. 69:704–716.

571 Bertels F., Silander O.K., Pachkov M., Rainey P.B., van Nimwegen E. 2014. Automated
572 Reconstruction of Whole-Genome Phylogenies from Short-Sequence Reads. *Molecular*
573 *Biology and Evolution*. 31:1077–1088.

574 Bohling J. 2020. Evaluating the effect of reference genome divergence on the analysis of
575 empirical RADseq datasets. *Ecology and Evolution*. 10:7585–7601.

576 Brandt D.Y.C., Aguiar V.R.C., Bitarello B.D., Nunes K., Goudet J., Meyer D. 2015. Mapping
577 Bias Overestimates Reference Allele Frequencies at the HLA Genes in the 1000
578 Genomes Project Phase I Data. *G3 Genes|Genomes|Genetics*. 5:931–941.

579 Bryant D., Hahn M.W. 2020. The Concatenation Question. In: Scornavacca C., Delsuc F.,
580 Galtier N., editors. *Phylogenetics in the Genomic Era*. No commercial publisher |
581 Authors open access book. p. 3.4:1-3.4:23.

582 Camacho C., Coulouris G., Avagyan V., Ma N., Papadopoulos J., Bealer K., Madden T.L.
583 2009. BLAST+: architecture and applications. *BMC Bioinformatics*. 10:421.

584 Capella-Gutiérrez S., Silla-Martínez J.M., Gabaldón T. 2009. trimAl: a tool for automated
585 alignment trimming in large-scale phylogenetic analyses. *Bioinformatics*. 25:1972–
586 1973.

587 Chen N.-C., Solomon B., Mun T., Iyer S., Langmead B. 2021. Reference flow: reducing
588 reference bias using multiple population genomes. *Genome Biology*. 22:8.

589 Coombes B., Lux T., Akhunov E., Hall A. 2024. Introgressions lead to reference bias in wheat
590 RNA-seq analysis. *BMC Biology*. 22:56.

PHYLOGENETIC METHOD TO DETECT REFERENCE BIAS

591 Croucher N.J., Harris S.R., Fraser C., Quail M.A., Burton J., van der Linden M., McGee L.,
592 von Gottberg A., Song J.H., Ko K.S., Pichon B., Baker S., Parry C.M., Lambertsen
593 L.M., Shahinas D., Pillai D.R., Mitchell T.J., Dougan G., Tomasz A., Klugman K.P.,
594 Parkhill J., Hanage W.P., Bentley S.D. 2011. Rapid Pneumococcal Evolution in
595 Response to Clinical Interventions. *Science*. 331:430–434.

596 Degnan J.H., Rosenberg N.A. 2009. Gene tree discordance, phylogenetic inference and the
597 multispecies coalescent. *Trends Ecol Evol*. 24:332–340.

598 Degner J.F., Marioni J.C., Pai A.A., Pickrell J.K., Nkadori E., Gilad Y., Pritchard J.K. 2009.
599 Effect of read-mapping biases on detecting allele-specific expression from RNA-
600 sequencing data. *Bioinformatics*. 25:3207–3212.

601 Dolenz S., van der Valk T., Jin C., Oppenheimer J., Sharif M.B., Orlando L., Shapiro B., Dalén
602 L., Heintzman P.D. 2024. Unravelling reference bias in ancient DNA datasets.
603 *Bioinformatics*. 40:btac436.

604 Dutilh B.E., Huynen M.A., Strous M. 2009. Increasing the coverage of a metapopulation
605 consensus genome by iterative read mapping and assembly. *Bioinformatics*. 25:2878–
606 2881.

607 Eddy S.R. 2011. Accelerated Profile HMM Searches. *PLoS Comput Biol*. 7:e1002195.

608 Edelman N.B., Frandsen P.B., Miyagi M., Clavijo B., Davey J., Dikow R.B., García-Accinelli
609 G., Van Belleghem S.M., Patterson N., Neafsey D.E., Challis R., Kumar S., Moreira
610 G.R.P., Salazar C., Chouteau M., Counterman B.A., Papa R., Blaxter M., Reed R.D.,
611 Dasmahapatra K.K., Kronforst M., Joron M., Jiggins C.D., McMillan W.O., Di Palma
612 F., Blumberg A.J., Wakeley J., Jaffe D., Mallet J. 2019. Genomic architecture and
613 introgression shape a butterfly radiation. *Science*. 366:594–599.

614 Ekblom R., Wolf J.B.W. 2014. A field guide to whole-genome sequencing, assembly and
615 annotation. *Evolutionary Applications*. 7:1026–1042.

Ivan et al.

- 616 Feigin C.Y., Newton A.H., Doronina L., Schmitz J., Hipsley C.A., Mitchell K.J., Gower G.,
617 Llamas B., Soubrier J., Heider T.N., Menzies B.R., Cooper A., O'Neill R.J., Pask A.J.
618 2018. Genome of the Tasmanian tiger provides insights into the evolution and
619 demography of an extinct marsupial carnivore. *Nat Ecol Evol.* 2:182–192.
- 620 Ferguson S., Jones A., Murray K., Andrew R., Schwessinger B., Borevitz J. 2024a. Plant
621 genome evolution in the genus *Eucalyptus* is driven by structural rearrangements that
622 promote sequence divergence. *Genome Res.* 34:606–619.
- 623 Ferguson S., Jones A., Murray K., Andrew R.L., Schwessinger B., Bothwell H., Borevitz J.
624 2024b. Exploring the role of polymorphic interspecies structural variants in
625 reproductive isolation and adaptive divergence in *Eucalyptus*. *GigaScience*.
626 13:giae029.
- 627 Gadagkar S.R., Rosenberg M.S., Kumar S. 2005. Inferring species phylogenies from multiple
628 genes: Concatenated sequence tree versus consensus gene tree. *Journal of Experimental*
629 *Zoology Part B: Molecular and Developmental Evolution.* 304B:64–74.
- 630 Gopalakrishnan S., Samaniego Castruita J.A., Sinding M.-H.S., Kuderna L.F.K., Räikkönen J.,
631 Petersen B., Sicheritz-Ponten T., Larson G., Orlando L., Marques-Bonet T., Hansen
632 A.J., Dalén L., Gilbert M.T.P. 2017. The wolf reference genome sequence (*Canis lupus*
633 *lupus*) and its implications for *Canis* spp. population genomics. *BMC Genomics*.
634 18:495.
- 635 Günther T., Nettelblad C. 2019. The presence and impact of reference bias on population
636 genomic studies of prehistoric human populations. *PLOS Genetics.* 15:e1008302.
- 637 Hahn C., Bachmann L., Chevreux B. 2013. Reconstructing mitochondrial genomes directly
638 from genomic next-generation sequencing reads—a baiting and iterative mapping
639 approach. *Nucleic Acids Research.* 41:e129.

PHYLOGENETIC METHOD TO DETECT REFERENCE BIAS

640 He M.-Q., Cao B., Liu F., Boekhout T., Denchev T.T., Schoutteten N., Denchev C.M., Kemler
641 M., Gorjón S.P., Begerow D., Valenzuela R., Davoodian N., Niskanen T., Vizzini A.,
642 Redhead S.A., Ramírez-Cruz V., Papp V., Dudka V.A., Dutta A.K., García-Sandoval
643 R., Liu X.-Z., Kijpornyongpan T., Savchenko A., Tedersoo L., Theelen B., Trierveiler-
644 Pereira L., Wu F., Zamora J.C., Zeng X.-Y., Zhou L.-W., Liu S.-L., Ghobad-Nejhad
645 M., Giachini A.J., Li G.-J., Kakishima M., Olariaga I., Haelewaters D., Sulistyo B.,
646 Sugiyama J., Svantesson S., Yurkov A., Alvarado P., Antonín V., da Silva A.F.,
647 Druzhinina I., Gibertoni T.B., Guzmán-Dávalos L., Justo A., Karunarathna S.C.,
648 Galappaththi M.C.A., Toome-Heller M., Hosoya T., Liimatainen K., Márquez R.,
649 Mešić A., Moncalvo J.-M., Nagy L.G., Varga T., Orihara T., Raymundo T., Salcedo I.,
650 Silva-Filho A.G.S., Tkalčec Z., Wartchow F., Zhao C.-L., Bau T., Cabarroi-Hernández
651 M., Cortés-Pérez A., Decock C., De Lange R., Weiss M., Menolli N., Nilsson R.H.,
652 Fan Y.-G., Verbeken A., Gafforov Y., Meiras-Otoni A., Mendes-Alvarenga R.L., Zeng
653 N.-K., Wu Q., Hyde K.D., Kirk P.M., Zhao R.-L. 2024. Phylogenomics, divergence
654 times and notes of orders in Basidiomycota. *Fungal Diversity*. 126:127–406.

655 Hoang D.T., Chernomor O., von Haeseler A., Minh B.Q., Vinh L.S. 2018. UFBoot2:
656 Improving the Ultrafast Bootstrap Approximation. *Molecular Biology and Evolution*.
657 35:518–522.

658 Homer N., Nelson S.F. 2010. Improved variant discovery through local re-alignment of short-
659 read next-generation sequencing data using SRMA. *Genome Biology*. 11:R99.

660 Huang P.-H., Wang T.-R., Li M., Fang O.-Y., Su R.-P., Meng H.-H., Song Y.-G., Li J. 2024.
661 Different reference genomes determine different results: Comparing SNP calling in
662 RAD-seq of *Engelhardia roxburghiana* using different reference genomes. *Plant*
663 *Science*. 344:112109.

Ivan et al.

664 Johnson M.G., Gardner E.M., Liu Y., Medina R., Goffinet B., Shaw A.J., Zerega N.J.C.,
665 Wickett N.J. 2016. HybPiper: Extracting coding sequence and introns for phylogenetics
666 from high-throughput sequencing reads using target enrichment1. *Appl Plant Sci.*
667 4:apps.1600016.

668 Kalyaanamoorthy S., Minh B.Q., Wong T.K.F., von Haeseler A., Jermin L.S. 2017.
669 ModelFinder: fast model selection for accurate phylogenetic estimates. *Nat Methods.*
670 14:587–589.

671 Kanzi A.M., Trollip C., Wingfield M.J., Barnes I., Van der Nest M.A., Wingfield B.D. 2020.
672 Phylogenomic incongruence in *Ceratocystis*: a clue to speciation? *BMC Genomics.*
673 21:362.

674 Katoh K., Misawa K., Kuma K., Miyata T. 2002. MAFFT: a novel method for rapid multiple
675 sequence alignment based on fast Fourier transform. *Nucleic Acids Res.* 30:3059–3066.

676 Kim D., Paggi J.M., Park C., Bennett C., Salzberg S.L. 2019. Graph-based genome alignment
677 and genotyping with HISAT2 and HISAT-genotype. *Nat Biotechnol.* 37:907–915.

678 Lanfear R., Hahn M.W. 2024. The Meaning and Measure of Concordance Factors in
679 Phylogenomics. *Molecular Biology and Evolution.* 41:msae214.

680 Levy Karin E., Mirdita M., Söding J. 2020. MetaEuk—sensitive, high-throughput gene
681 discovery, and annotation for large-scale eukaryotic metagenomics. *Microbiome.* 8:48.

682 Li H. 2011. A statistical framework for SNP calling, mutation discovery, association mapping
683 and population genetical parameter estimation from sequencing data. *Bioinformatics.*
684 27:2987–2993.

685 Li H. 2023. Protein-to-genome alignment with minimap2. *Bioinformatics.* 39:btad014.

686 Li H., Durbin R. 2009. Fast and accurate short read alignment with Burrows–Wheeler
687 transform. *Bioinformatics.* 25:1754–1760.

PHYLOGENETIC METHOD TO DETECT REFERENCE BIAS

688 Li H., Handsaker B., Wysoker A., Fennell T., Ruan J., Homer N., Marth G., Abecasis G.,
689 Durbin R., 1000 Genome Project Data Processing Subgroup. 2009. The Sequence
690 Alignment/Map format and SAMtools. *Bioinformatics*. 25:2078–2079.

691 Li R., Fan W., Tian G., Zhu H., He L., Cai J., Huang Q., Cai Q., Li B., Bai Y., Zhang Z., Zhang
692 Y., Wang W., Li J., Wei F., Li H., Jian M., Li J., Zhang Z., Nielsen R., Li D., Gu W.,
693 Yang Z., Xuan Z., Ryder O.A., Leung F.C.-C., Zhou Y., Cao J., Sun X., Fu Y., Fang
694 X., Guo X., Wang B., Hou R., Shen F., Mu B., Ni P., Lin R., Qian W., Wang G., Yu
695 C., Nie W., Wang J., Wu Z., Liang H., Min J., Wu Q., Cheng S., Ruan J., Wang M.,
696 Shi Z., Wen M., Liu B., Ren X., Zheng H., Dong D., Cook K., Shan G., Zhang H.,
697 Kosiol C., Xie X., Lu Z., Zheng H., Li Y., Steiner C.C., Lam T.T.-Y., Lin S., Zhang
698 Q., Li G., Tian J., Gong T., Liu H., Zhang D., Fang L., Ye C., Zhang J., Hu W., Xu A.,
699 Ren Y., Zhang G., Bruford M.W., Li Q., Ma L., Guo Y., An N., Hu Y., Zheng Y., Shi
700 Y., Li Z., Liu Q., Chen Y., Zhao J., Qu N., Zhao S., Tian F., Wang X., Wang H., Xu
701 L., Liu X., Vinar T., Wang Y., Lam T.-W., Yiu S.-M., Liu S., Zhang H., Li D., Huang
702 Y., Wang X., Yang G., Jiang Z., Wang J., Qin N., Li L., Li J., Bolund L., Kristiansen
703 K., Wong G.K.-S., Olson M., Zhang X., Li S., Yang H., Wang J., Wang J. 2010. The
704 sequence and de novo assembly of the giant panda genome. *Nature*. 463:311–317.

705 Li Y., Steenwyk J.L., Chang Y., Wang Y., James T.Y., Stajich J.E., Spatafora J.W.,
706 Groenewald M., Dunn C.W., Hittinger C.T., Shen X.-X., Rokas A. 2021. A genome-
707 scale phylogeny of the kingdom Fungi. *Current Biology*. 31:1653-1665.e5.

708 Liao X., Li M., Zou Y., Wu F.-X., Yi-Pan, Wang J. 2019. Current challenges and solutions of
709 de novo assembly. *Quantitative Biology*. 7:90–109.

710 Lin M.-J., Iyer S., Chen N.-C., Langmead B. 2024. Measuring, visualizing, and diagnosing
711 reference bias with biastools. *Genome Biology*. 25:101.

Ivan et al.

- 712 Lozano-Fernandez J. 2022. A Practical Guide to Design and Assess a Phylogenomic Study.
713 Genome Biology and Evolution. 14:evac129.
- 714 Mallet J., Besansky N., Hahn M.W. 2016. How reticulated are species? BioEssays. 38:140–
715 149.
- 716 Manni M., Berkeley M.R., Seppey M., Zdobnov E.M. 2021. BUSCO: Assessing Genomic Data
717 Quality and Beyond. Current Protocols. 1:e323.
- 718 Minh B.Q., Hahn M.W., Lanfear R. 2020a. New Methods to Calculate Concordance Factors
719 for Phylogenomic Datasets. Mol Biol Evol. 37:2727–2733.
- 720 Minh B.Q., Schmidt H.A., Chernomor O., Schrempf D., Woodhams M.D., von Haeseler A.,
721 Lanfear R. 2020b. IQ-TREE 2: New Models and Efficient Methods for Phylogenetic
722 Inference in the Genomic Era. Molecular Biology and Evolution. 37:1530–1534.
- 723 Mo Y.K., Lanfear R., Hahn M.W., Minh B.Q. 2023. Updated site concordance factors
724 minimize effects of homoplasy and taxon sampling. Bioinformatics. 39:btac741.
- 725 Mose L.E., Perou C.M., Parker J.S. 2019. Improved indel detection in DNA and RNA via
726 realignment with ABRA2. Bioinformatics. 35:2966–2973.
- 727 Munger S.C., Raghupathy N., Choi K., Simons A.K., Gatti D.M., Hinerfeld D.A., Svenson
728 K.L., Keller M.P., Attie A.D., Hibbs M.A., Graber J.H., Chesler E.J., Churchill G.A.
729 2014. RNA-Seq Alignment to Individualized Genomes Improves Transcript
730 Abundance Estimates in Multiparent Populations. Genetics. 198:59–73.
- 731 Neph S., Kuehn M.S., Reynolds A.P., Haugen E., Thurman R.E., Johnson A.K., Rynes E.,
732 Maurano M.T., Vierstra J., Thomas S., Sandstrom R., Humbert R.,
733 Stamatoyannopoulos J.A. 2012. BEDOPS: high-performance genomic feature
734 operations. Bioinformatics. 28:1919–1920.
- 735 Nevado B., Ramos-Onsins S.E., Perez-Enciso M. 2014. Resequencing studies of nonmodel
736 organisms using closely related reference genomes: optimal experimental designs and

PHYLOGENETIC METHOD TO DETECT REFERENCE BIAS

737 bioinformatics approaches for population genomics. *Molecular Ecology*. 23:1764–
738 1779.

739 Ondov B.D., Treangen T.J., Melsted P., Mallonee A.B., Bergman N.H., Koren S., Phillippy
740 A.M. 2016. Mash: fast genome and metagenome distance estimation using MinHash.
741 *Genome Biology*. 17:132.

742 Panousis N.I., Gutierrez-Arcelus M., Dermitzakis E.T., Lappalainen T. 2014. Allelic mapping
743 bias in RNA-sequencing is not a major confounder in eQTL studies. *Genome Biology*.
744 15:467.

745 Paradis E., Claude J., Strimmer K. 2004. APE: Analyses of Phylogenetics and Evolution in R
746 language. *Bioinformatics*. 20:289–290.

747 Paszkiewicz K., Studholme D.J. 2010. De novo assembly of short sequence reads. *Briefings in*
748 *Bioinformatics*. 11:457–472.

749 Pease J.B., Haak D.C., Hahn M.W., Moyle L.C. 2016. Phylogenomics Reveals Three Sources
750 of Adaptive Variation during a Rapid Radiation. *PLOS Biology*. 14:e1002379.

751 Pirooznia M., Kramer M., Parla J., Goes F.S., Potash J.B., McCombie W.R., Zandi P.P. 2014.
752 Validation and assessment of variant calling pipelines for next-generation sequencing.
753 *Hum Genomics*. 8:14.

754 R Core Team. 2023. R: A language and environment for statistical computing. .

755 Rabiee M., Mirarab S. 2020. Forcing external constraints on tree inference using ASTRAL.
756 *BMC Genomics*. 21:218.

757 Rhie A., McCarthy S.A., Fedrigo O., Damas J., Formenti G., Koren S., Uliano-Silva M., Chow
758 W., Fungtammasan A., Kim J., Lee C., Ko B.J., Chaisson M., Gedman G.L., Cantin
759 L.J., Thibaud-Nissen F., Haggerty L., Bista I., Smith M., Haase B., Mountcastle J.,
760 Winkler S., Paez S., Howard J., Vernes S.C., Lama T.M., Grutzner F., Warren W.C.,
761 Balakrishnan C.N., Burt D., George J.M., Biegler M.T., Iorns D., Digby A., Eason D.,

Ivan et al.

Robertson B., Edwards T., Wilkinson M., Turner G., Meyer A., Kautt A.F., Franchini P., Detrich H.W., Svardal H., Wagner M., Naylor G.J.P., Pippel M., Malinsky M., Mooney M., Simbirsky M., Hannigan B.T., Pesout T., Houck M., Misuraca A., Kingan S.B., Hall R., Kronenberg Z., Sović I., Dunn C., Ning Z., Hastie A., Lee J., Selvaraj S., Green R.E., Putnam N.H., Gut I., Ghurye J., Garrison E., Sims Y., Collins J., Pelan S., Torrance J., Tracey A., Wood J., Dagnew R.E., Guan D., London S.E., Clayton D.F., Mello C.V., Friedrich S.R., Lovell P.V., Osipova E., Al-Ajli F.O., Secomandi S., Kim H., Theofanopoulou C., Hiller M., Zhou Y., Harris R.S., Makova K.D., Medvedev P., Hoffman J., Masterson P., Clark K., Martin F., Howe K., Flicek P., Walenz B.P., Kwak W., Clawson H., Diekhans M., Nassar L., Paten B., Kraus R.H.S., Crawford A.J., Gilbert M.T.P., Zhang G., Venkatesh B., Murphy R.W., Koepfli K.-P., Shapiro B., Johnson W.E., Di Palma F., Marques-Bonet T., Teeling E.C., Warnow T., Graves J.M., Ryder O.A., Haussler D., O'Brien S.J., Korlach J., Lewin H.A., Howe K., Myers E.W., Durbin R., Phillippy A.M., Jarvis E.D. 2021. Towards complete and error-free genome assemblies of all vertebrate species. *Nature*. 592:737–746.

Rick J.A., Brock C.D., Lewanski A.L., Golcher-Benavides J., Wagner C.E. 2024. Reference Genome Choice and Filtering Thresholds Jointly Influence Phylogenomic Analyses. *Systematic Biology*. 73:76–101.

Schneeberger K., Hagmann J., Ossowski S., Warthmann N., Gesing S., Kohlbacher O., Weigel D. 2009. Simultaneous alignment of short reads against multiple genomes. *Genome Biology*. 10:R98.

Schober P., Boer C., Schwarte L.A. 2018. Correlation Coefficients: Appropriate Use and Interpretation. *Anesthesia & Analgesia*. 126:1763.

Schubert M., Lindgreen S., Orlando L. 2016. AdapterRemoval v2: rapid adapter trimming, identification, and read merging. *BMC Research Notes*. 9:88.

PHYLOGENETIC METHOD TO DETECT REFERENCE BIAS

787 Sedlazeck F.J., Lee H., Darby C.A., Schatz M.C. 2018. Piercing the dark matter: bioinformatics
788 of long-range sequencing and mapping. *Nat Rev Genet.* 19:329–346.

789 Shafer A.B.A., Peart C.R., Tusso S., Maayan I., Brelsford A., Wheat C.W., Wolf J.B.W. 2017.
790 Bioinformatic processing of RAD-seq data dramatically impacts downstream
791 population genetic inference. *Methods in Ecology and Evolution.* 8:907–917.

792 Simão F.A., Waterhouse R.M., Ioannidis P., Kriventseva E.V., Zdobnov E.M. 2015. BUSCO:
793 assessing genome assembly and annotation completeness with single-copy orthologs.
794 *Bioinformatics.* 31:3210–3212.

795 Sirén J., Monlong J., Chang X., Novak A.M., Eizenga J.M., Markello C., Sibbesen J.A., Hickey
796 G., Chang P.-C., Carroll A., Gupta N., Gabriel S., Blackwell T.W., Ratan A., Taylor
797 K.D., Rich S.S., Rotter J.I., Haussler D., Garrison E., Paten B. 2021. Pangenomics
798 enables genotyping of known structural variants in 5202 diverse genomes. *Science.*
799 374:abg8871.

800 Stanke M., Steinkamp R., Waack S., Morgenstern B. 2004. AUGUSTUS: a web server for
801 gene finding in eukaryotes. *Nucleic Acids Res.* 32:W309–W312.

802 Stevenson K.R., Coolon J.D., Wittkopp P.J. 2013. Sources of bias in measures of allele-specific
803 expression derived from RNA-seq data aligned to a single reference genome. *BMC*
804 *Genomics.* 14:536.

805 Stiller J., Feng S., Chowdhury A.-A., Rivas-González I., Duchêne D.A., Fang Q., Deng Y.,
806 Kozlov A., Stamatakis A., Claramunt S., Nguyen J.M.T., Ho S.Y.W., Faircloth B.C.,
807 Haag J., Houde P., Cracraft J., Balaban M., Mai U., Chen G., Gao R., Zhou C., Xie Y.,
808 Huang Z., Cao Z., Yan Z., Ogilvie H.A., Nakhleh L., Lindow B., Morel B., Fjeldså J.,
809 Hosner P.A., da Fonseca R.R., Petersen B., Tobias J.A., Székely T., Kennedy J.D.,
810 Reeve A.H., Liker A., Stervander M., Antunes A., Tietze D.T., Bertelsen M.F., Lei F.,
811 Rahbek C., Graves G.R., Schierup M.H., Warnow T., Braun E.L., Gilbert M.T.P., Jarvis

Ivan et al.

- 812 E.D., Mirarab S., Zhang G. 2024. Complexity of avian evolution revealed by family-
813 level genomes. *Nature*. 629:851–860.
- 814 The Computational Pan-Genomics Consortium. 2018. Computational pan-genomics: status,
815 promises and challenges. *Briefings in Bioinformatics*. 19:118–135.
- 816 Tsai I.J., Otto T.D., Berriman M. 2010. Improving draft assemblies by iterative mapping and
817 assembly of short reads to eliminate gaps. *Genome Biol*. 11:R41.
- 818 Valenzuela D., Norri T., Välimäki N., Pitkänen E., Mäkinen V. 2018. Towards pan-genome
819 read alignment to improve variation calling. *BMC Genomics*. 19:87.
- 820 Valiente-Mullor C., Beamud B., Ansari I., Francés-Cuesta C., García-González N., Mejía L.,
821 Ruiz-Hueso P., González-Candelas F. 2021. One is not enough: On the effects of
822 reference genome for the mapping and subsequent analyses of short-reads. *PLOS*
823 *Computational Biology*. 17:e1008678.
- 824 Varga T., Woods R., Pitsillides F., Hill R., Biketova A.Y., Llewellyn T., Shaw B.J.P., Hodgson
825 E., Wong B., Le J., Becker J., Bradshaw A.J., Blake S.L.E., Geeves C., Levicky Q.,
826 Goodman L., Mole R.L., Reed S.L., Lin M.C., Read E., Harris K., Pino-Bodas R., Nagy
827 L.G., Bazzicalupo A., Gaya E. 2025. Whole genome sequencing of historical specimens
828 from the world’s largest fungal collection yields high-quality assemblies. *New*
829 *Phytologist*. n/a.
- 830 Vasimuddin M., Misra S., Li H., Aluru S. 2019. Efficient Architecture-Aware Acceleration of
831 BWA-MEM for Multicore Systems. 2019 IEEE International Parallel and Distributed
832 Processing Symposium (IPDPS):314–324.
- 833 Young A.D., Gillung J.P. 2020. Phylogenomics — principles, opportunities and pitfalls of big-
834 data phylogenetics. *Systematic Entomology*. 45:225–247.

PHYLOGENETIC METHOD TO DETECT REFERENCE BIAS

- 835 Zdobnov E.M., Kuznetsov D., Tegenfeldt F., Manni M., Berkeley M., Kriventseva E.V. 2021.
836 OrthoDB in 2020: evolutionary and functional annotations of orthologs. *Nucleic Acids*
837 *Research*. 49:D389–D393.
- 838 Zhan S., Griswold C., Lukens L. 2021. Zea mays RNA-seq estimated transcript abundances
839 are strongly affected by read mapping bias. *BMC Genomics*. 22:285.
- 840 Zhang C., Rabiee M., Sayyari E., Mirarab S. 2018. ASTRAL-III: polynomial time species tree
841 reconstruction from partially resolved gene trees. *BMC Bioinformatics*. 19:153.
- 842